

**Spain survey:
19,295 diaries**

**Italy survey:
41,229 diaries**

**United Kingdom
survey:
16,533 diaries**

**Harmonise to
shared activity
codes**

**Convert
diaries to
text records**

**Train on two
countries**

**Fine-tuned model generates
held-out diaries**

**Reweight real diaries
by IPF**

**Compare to
published
time budgets**

**Pre-registered
pass or fail rule**

**Diaries drive
building
simulation**

**Occupancy-based
internal gains**

**Activity-triggered
appliance loads**

**Domestic hot
water draws**

**Heating demand:
Spain, Italy, UK archetypes**

**Stock-scale loads:
London and Bologna**